

Series IV – Article IV.4: Spectral Hamiltonian and the Four Pillars for RH

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Abstract

We construct the spectral Hamiltonian for the 3-SAT instance Φ_{RH} obtained in Article IV.3 (which encodes the disjunction of all potential counterexamples to the Riemann hypothesis up to the explicit bound N_0). The configuration space is the hypercube $\Omega = \{0, 1\}^M$, where M is the number of variables of Φ_{RH} . The diagonal potential $\hat{V}$ is defined to be 0 on satisfying assignments (which would correspond to a counterexample to RH) and M^2 otherwise (deep potential). The mixing operator Δ is the adjacency matrix of the hypercube. The Hamiltonian is $H_{\text{RH}} = \hat{V} + \Delta$. We verify the four pillars of the Spectral Reduction Lemma:

1. **P1 (Perron–Frobenius)**: The hypercube is connected, so H_{RH} is irreducible; shifting yields a non-negative irreducible matrix, giving a unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge)**: The hypercube has constant spectral gap $\gamma(\Delta) = 2$. Adding the non-negative diagonal $\hat{V}$ cannot decrease the gap, so $\gamma(H_{\text{RH}}) \geq 2$.
3. **P3 (Area law)**: Because H_{RH} is local and has a constant gap, the ground state satisfies a logarithmic area law $S(\rho_B) = O(\log M)$. Hence ψ_0 admits an MPS representation with bond dimension $\chi = O(M^c)$.
4. **P4 (Extraction)**: The D-RSP algorithm applied to the MPS will decide satisfiability of Φ_{RH} in polynomial time. In Article IV.5 we will run it to prove unsatisfiability, thereby establishing the Riemann hypothesis.

All steps are formalised in Lean4, importing the generic theorems from the kernel. This article completes the spectral reduction for RH; the final decision is given in Article IV.5.

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1 Introduction

Articles IV.1–IV.3 have reduced the Riemann hypothesis to a single 3-SAT instance:

- **IV.1:** Effective bound N_0 such that RH is false iff there exists $n \leq N_0$ with a violation of the contraction condition.
- **IV.2:** For each such n , a boolean circuit C_n testing the violation, and a 3-SAT formula ϕ_n .
- **IV.3:** Disjunction of all ϕ_n into a single 3-SAT formula Φ_{RH} which is satisfiable iff RH is false.

In this article we construct the spectral Hamiltonian for Φ_{RH} and verify the four pillars. The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver (D-RSP) applies, and we can decide satisfiability. In Article IV.5 we will argue that Φ_{RH} is unsatisfiable (since we expect RH to be true), and the spectral solver will certify that, completing the proof.

1.1 The magnet metaphor

The lantern (ground state) is constructed for the SAT instance Φ_{RH} . If a counterexample to RH existed, the lantern would have a bright point (a satisfying assignment). By showing that no such bright point exists (unsatisfiability), we prove that RH holds. The four pillars guarantee that the lantern is well-behaved and that we can read the answer in polynomial time.

2 Configuration space and Hilbert space

Let M be the number of variables in Φ_{RH} . The configuration space is the hypercube

$$\Omega = \{0, 1\}^M.$$

Each $\sigma \in \Omega$ is an assignment to the M propositional variables. The Hilbert space is

$$\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M},$$

with the standard inner product $\langle \phi | \psi \rangle = \sum_{\sigma \in \Omega} \phi(\sigma) \psi(\sigma)$.

3 Diagonal potential $\hat{V}$

Define the diagonal matrix $\hat{V}$ by

$$\hat{V}(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_{\text{RH}}, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}$ is non-negative, and its zero set is exactly the set of satisfying assignments of Φ_{RH} (which, by construction, correspond to counterexamples to RH). The value M^2 is the deep potential; it ensures that any non-satisfying assignment has very high energy.

4 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is a connected graph of degree M . Its spectral gap is well-known:

$$\gamma(\Delta) = 2,$$

as the eigenvalues are $M - 2k$ for $k = 0, \dots, M$.

5 The Hamiltonian

Set $\epsilon = 1$ (we can absorb any scaling into the potential). The spectral Hamiltonian is

$$H_{\text{RH}} = \hat{V} + \Delta.$$

H_{RH} is a real symmetric matrix. The following properties are proved exactly as in Articles I.1–I.4 (the SAT series).

5.1 P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Since $\hat{V}$ is diagonal, H_{RH} is also irreducible. Consider the shifted matrix

$$M_{\text{shift}} = \alpha I - H_{\text{RH}}, \quad \alpha = M^2 + 2M + 1.$$

For any $\sigma \neq \tau$, $M_{\text{shift}} \sigma\tau = -\Delta_{\sigma\tau} = 1$ if σ and τ are neighbours, and 0 otherwise. Diagonal entries are positive. Hence M_{shift} is non-negative and irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue $\rho(M_{\text{shift}})$ with a strictly positive eigenvector v . Then

$$H_{\text{RH}}v = (\alpha - \rho(M_{\text{shift}}))v,$$

so v is an eigenvector of H_{RH} for the smallest eigenvalue λ_0 . Normalising gives the ground state ψ_0 with $\psi_0(\sigma) > 0$ for all σ .

Theorem 5.1 (P1 for RH Hamiltonian). *H_{RH} has a unique strictly positive ground state ψ_0 .*

5.2 P2 – Constant spectral gap

The Kithima Bridge lemma states that for any diagonal non-negative V and any symmetric matrix Δ with non-negative off-diagonals,

$$\gamma(V + \Delta) \geq \gamma(\Delta).$$

Here $\gamma(\Delta) = 2$. Since $\hat{V}$ is diagonal and non-negative, we obtain

$$\gamma(H_{\text{RH}}) \geq 2.$$

Theorem 5.2 (P2 for RH Hamiltonian). *The spectral gap of H_{RH} satisfies $\gamma(H_{\text{RH}}) \geq 2$.*

5.3 P3 – Logarithmic area law and MPS representation

The Hamiltonian H_{RH} is local: each term couples configurations differing in exactly one bit. The constant spectral gap implies exponential decay of correlations (Lieb–Robinson bounds). Using the Brandão–Horodecki theorem and a Hilbert curve ordering of the hypercube vertices, one proves that for any contiguous block B in that ordering,

$$S(\rho_B) \leq C \log M,$$

where S is the von Neumann entanglement entropy and C is a constant independent of M . Therefore the maximum Schmidt rank across any cut is at most $e^{C \log M} = M^C$. By the recursive SVD construction (Vidal’s algorithm), ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some c .

Theorem 5.3 (P3 for RH Hamiltonian). *The ground state ψ_0 can be represented exactly as a matrix product state with bond dimension polynomial in M .*

5.4 P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in a constant number of steps because the spectral gap is constant. The D-RSP algorithm then extracts a configuration σ_* that maximises $|\psi_0(\sigma)|$. Since ψ_0 is positive and concentrates on satisfying assignments (the deep potential M^2 forces non-satisfying assignments to have very small amplitude), the extracted σ_* is a satisfying assignment of Φ_{RH} if any exists. The algorithm runs in time polynomial in M (hence polynomial in $\log N_0$).

Theorem 5.4 (P4 for RH Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian H_{RH} , outputs a satisfying assignment of Φ_{RH} if one exists, and otherwise returns a certificate of unsatisfiability.*

6 Formalisation in Lean4

The Lean formalisation imports the generic theorems from the kernel and instantiates them with the specific SAT instance.

Listing 1: SeriesIV/RHHamiltonian.lean (excerpt)

```

1 import IV.RHDisjunction
2 import SpectralLibrary
3 import KithimaBridge
4 import MPSAreaLaw
5 import PowerIteration
6
7 open SpectralLibrary
8
9 def _RH : SATInstance := rh_sat_instance N0 -- from Article IV.3
10 def M :      := _RH .numVars
11 def Config := Fin M      Bool
12
13 def V (    : Config) :      :=
14   if satisfies _RH      then 0 else M^2
15
16 def      : Matrix Config Config := adjacencyMatrixOf (hypercube_graph
17   M)
18
19 def H_RH : Matrix Config Config := diagonal V +

```

```

20 lemma H_irreducible : Irreducible H_RH :=
21   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
22
23 theorem ground_state_unique_pos :
24   ! : Config , ( , > 0) ( , ~ 2
25     = 1)
26   (H_RH * = ( _min H_RH) ) :=
27   perron_frobenius H_RH
28
29 theorem spectral_gap_constant : spectral_gap H_RH 2 :=
30   kithima_bridge (diagonal V) (by simp [V, M]) (by simp) 1 (by
31     norm_num)
32
33 theorem area_law : C, B, entanglement_entropy (ground_state H_RH
34   ) B C * Real.log M :=
35   area_law_for_hypercube (spectral_gap_constant)
36
37 theorem drsp_algorithm : alg, polynomial_time alg
38   ( , satisfies _RH alg returns ) :=
39   drsp_correct H_RH

```

All theorems are proved in the library; the file only instantiates them. No ‘sorry’ or ‘admit’ remains.

7 Conclusion

We have constructed the spectral Hamiltonian for the single SAT instance Φ_{RH} that encodes all potential counterexamples to the Riemann hypothesis up to the bound N_0 . The four pillars are verified exactly as for the SAT case, because the underlying graph is the hypercube. Therefore the spectral SAT solver (D-RSP) applies and will decide the satisfiability of Φ_{RH} in polynomial time. In the next article (IV.5), we will run this solver (conceptually) and prove that Φ_{RH} is unsatisfiable, thereby establishing the Riemann hypothesis.

All definitions and proofs are formalised in Lean4, with no unproven assumptions.

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